Context-Window Degradation and Its Impact on Long Document Research Synthesis

Abstract

The rapid expansion of large language model (LLM) context windows has created the impression that increasingly long scholarly documents can be processed as unified inputs without substantial loss of analytical quality. However, a growing body of research demonstrates a distinction between **context-window capacity**—the maximum amount of text a model can accept—and **effective long-context utilization**—the extent to which the model can reliably retrieve, integrate, reason over, and synthesize information distributed across that context. This paper examines *context-window degradation*, defined here as the decline in effective task performance as contextual length increases, even when the input remains within the model's nominal context limit. Particular attention is given to its consequences for long-document research synthesis, where relevant evidence may be distributed across hundreds of pages, separated by irrelevant material, repeated across sources, or dependent on multi-hop relationships. The literature indicates several interacting mechanisms: positional biases, dilution by irrelevant information, attention and computational constraints, imperfect retrieval, long-range dependency failures, and errors introduced during compression or hierarchical summarization. Empirical findings such as the "lost in the middle" effect show that models can perform substantially worse when relevant evidence is located in the middle of long contexts, while more recent evidence suggests that context length itself can degrade performance even when retrieval is perfect. Current mitigation strategies include retrieval-augmented generation, prompt compression, document reordering, hierarchical synthesis, long-context pretraining, positional-encoding extension, external memory, and multi-stage reasoning. The paper argues that reliable scholarly synthesis should therefore be treated not as a single-pass long-context generation problem but as an evidence-management and iterative reasoning problem. It concludes by identifying research gaps involving evaluation of synthesis quality, provenance preservation, contradiction handling, uncertainty calibration, cross-document evidence integration, and domain-specific long-context benchmarks.

Keywords: large language models; long-context language models; context-window degradation; lost in the middle; long-document summarization; research synthesis; retrieval-augmented generation; prompt compression; scholarly information synthesis; evidence integration

1. Introduction

Large language models have rapidly evolved from systems designed primarily for relatively short prompts into models capable of accepting context windows ranging from tens of thousands to, in some systems, millions of tokens. This development has important implications for scholarly research. A researcher can potentially provide an entire monograph, a collection of research papers, a legal corpus, a technical report, or a large literature set to a model and request a synthesis. The apparent advantage is straightforward: instead of repeatedly retrieving and summarizing individual documents, the model can theoretically reason over the complete evidence base simultaneously.

Yet nominal context capacity should not be equated with reliable comprehension. The Transformer architecture underlying modern LLMs was originally designed around sequence processing in which self-attention has quadratic computational complexity with respect to sequence length (Vaswani et al., 2017). Although algorithmic innovations, optimized attention implementations, positional-encoding methods, and long-context training have substantially increased feasible sequence lengths, these advances do not automatically guarantee that a model will use all information in a long input effectively.

This distinction is especially consequential for **research synthesis**. Scholarly synthesis differs from simple summarization. It requires identifying relevant evidence, distinguishing primary from secondary claims, integrating findings across studies, reconciling methodological differences, detecting contradictions, preserving attribution, and deriving conclusions that are supported by the evidence. Relevant information may occur at arbitrary positions in documents and may only become meaningful when combined with information found elsewhere.

A seminal empirical study by Liu et al. (2024) demonstrated that language models can exhibit a pronounced positional bias in long contexts. In multi-document question answering and key-value retrieval, performance frequently followed a U-shaped pattern: models performed relatively well when relevant information appeared near the beginning or end of the context but substantially worse when the same information was placed in the middle. This phenomenon, commonly described as **"lost in the middle,"** challenges the assumption that a model possessing a sufficiently large context window can reliably exploit all information contained within it.

More recent research has strengthened this concern. Du et al. (2025) report that LLM performance can decline substantially as input length increases even when the relevant evidence is perfectly retrieved and distracting information is minimized or masked. Their results suggest that long-context degradation cannot be explained solely by retrieval failure.

The central argument of this paper is therefore that **long-document research synthesis should be understood as an effective-context problem rather than merely a context capacity problem**. Increasing the maximum number of tokens that a model can ingest is valuable, but it does not eliminate degradation in retrieval, attention allocation, reasoning, evidence integration, or synthesis. The paper reviews the relevant literature, analyzes mechanisms underlying degradation, surveys current mitigation approaches, and identifies directions for developing more reliable research-oriented long-context systems.

2. Background and Related Work

2.1 Transformer context processing

The Transformer architecture introduced by Vaswani et al. (2017) replaced recurrent sequence processing with self-attention, allowing models to establish direct interactions between tokens and enabling highly parallelized training. However, conventional full self attention has computational and memory requirements that scale quadratically with sequence length. This property historically constrained practical context windows.

FlashAttention demonstrated that substantial efficiency gains could be achieved without approximating the attention computation itself. Dao et al. (2022) proposed an I/O-aware exact-attention algorithm that reduces memory transfers between GPU memory levels, enabling substantially more efficient processing of longer sequences. Such engineering developments have been an important component of the transition toward long-context LLMs.

Nevertheless, efficient computation addresses only one dimension of long-context processing. A model can computationally process a sequence without reliably extracting and integrating its most important information.

2.2 Extending context windows

Long-context research has pursued several complementary strategies. One approach modifies positional representations so that models can generalize beyond their original training sequence length. LongRoPE, for example, uses non-uniform positional interpolation and progressive extension to extend pretrained models to contexts substantially longer than their original windows. Ding et al. (2024) report extensions reaching 2,048K tokens while attempting to preserve shorter-context performance.

LongT5 represents another strategy: Guo et al. (2022) integrated local/global attention ideas with the T5 architecture and pretraining approaches designed for summarization, demonstrating strong performance on long-document summarization and question answering.

These approaches establish an important principle: **context-window extension and long context competence are related but distinct objectives**. Increasing positional capacity can make longer inputs technically processable, but models must also learn to exploit those inputs effectively.

2.3 Retrieval-augmented generation

Retrieval-augmented generation (RAG) provides an alternative to placing an entire corpus into the model's context. Lewis et al. (2020) introduced a general framework that combines parametric model knowledge with an explicit non-parametric memory retrieved from an external corpus. Their experiments showed that RAG can improve knowledge intensive question answering while providing a mechanism for incorporating external evidence.

For research synthesis, retrieval can reduce the amount of irrelevant material presented to the generator. Instead of asking a model to process thousands of pages indiscriminately, a system can retrieve passages associated with a research question and then synthesize those passages.

However, retrieval introduces its own failure modes. Evidence can be omitted, ranked incorrectly, or fragmented across documents. Consequently, RAG does not eliminate context degradation; rather, it shifts part of the problem from **long-context utilization** to **evidence selection and context construction**.

2.4 Long-context evaluation

The development of dedicated benchmarks has made the distinction between nominal and effective context increasingly visible. LongBench introduced 21 datasets across six task categories, including single-document QA, multi-document QA, summarization, few shot learning, synthetic tasks, and code completion. Bai et al. (2024) found that models could still struggle on longer contexts even when their advertised context windows were sufficient to contain the inputs.

LooGLE similarly emphasizes long dependencies rather than simple context acceptance. Li et al. (2024) reported that representative LLMs frequently failed to capture long-range dependencies even when the document fit within the model's nominal context window.

These benchmarks reveal that "supports 128K tokens" or "supports 1M tokens" is not itself a meaningful measure of research-synthesis reliability. Evaluation must measure what the model can *do* with those tokens.

3. Context-Window Degradation

3.1 Definition

For the purposes of this paper, **context-window degradation (CWD)** is defined as a decline in task-relevant model performance associated with increasing contextual length, contextual dispersion, or contextual complexity, despite the input remaining within the model's stated context capacity.

This definition encompasses several related phenomena:

1. **Positional degradation:** relevant information becomes less accessible when it occurs at particular locations.
2. **Length-induced degradation:** performance declines as irrelevant or redundant context increases.
3. **Retrieval degradation:** relevant evidence is increasingly difficult to identify among competing material.
4. **Integration degradation:** the model retrieves individual facts but fails to combine them correctly.
5. **Synthesis degradation:** the model generates plausible prose while

omitting, misattributing, or contradicting evidence.

6. **Compression degradation:** summarization or context compression removes information necessary for later reasoning.

CWD is therefore multidimensional rather than a single model characteristic.

3.2 The "lost in the middle" effect

Liu et al. (2024) provide some of the strongest evidence for positional degradation. Their controlled experiments varied both the length of the context and the location of relevant information. Performance was often highest at the beginning and end and substantially lower in the middle. In some settings, GPT-3.5-Turbo's multi-document QA performance fell by more than 20 percentage points relative to favorable placements, with certain long context configurations performing worse than a closed-book baseline.

The implication for research synthesis is profound. In a 300-page literature corpus, a study's key methodological qualification might occur on page 173. If that information receives effectively less influence than material near the beginning or end of the constructed prompt, a synthesis can become systematically biased despite containing the correct evidence.

The problem is particularly serious because scholarly documents are structurally heterogeneous. Important information may appear in:

- methods sections;
- footnotes;
- tables;
- appendices;
- limitations;
- supplementary material;
- figure captions;
- discussion sections; or
- references to earlier studies.

Consequently, positional bias can interact with document structure to create systematic evidence-selection errors.

3.3 Length itself as a source of degradation

A second and potentially more fundamental problem is that increasing input length can reduce performance independently of retrieval.

Du et al. (2025) conducted controlled experiments in which relevant information was perfectly available and irrelevant material was minimized or masked. Nevertheless, performance declined as input length increased, with reported degradation ranging from 13.9% to 85% across tested tasks and models.

This finding changes the conceptual model of long-context failure. If degradation occurred only because models could not *find* the relevant passage, better retrieval would solve the problem. But if performance falls even after retrieval is perfect, then the model's subsequent reasoning process is itself sensitive to contextual scale.

For research synthesis, this means that simply retrieving the correct papers or passages may not be sufficient. A system must also construct a context that permits reliable comparison and reasoning.

3.4 Information dilution and interference

Long documents contain both signal and noise. Even when apparently relevant, many passages contribute little to a specific research question. Repetition, background information, methodological details, citations, and tangential discussions can compete for model attention.

LongLLMLingua explicitly frames long-context processing around three challenges: increased computational cost, performance reduction associated with irrelevant or redundant information, and positional bias. Jiang et al. (2024) report that question-aware prompt compression can improve performance while reducing token consumption and latency.

This suggests that the relationship between context length and performance is not necessarily monotonic in the useful sense. More context may provide more potentially relevant evidence while simultaneously increasing interference.

4. Impact on Long-Document Research Synthesis

4.1 Evidence retrieval

The first stage of scholarly synthesis is locating evidence relevant to the research question. Long-context degradation can cause models to overlook relevant passages or disproportionately weight evidence based on position.
This creates a subtle failure mode: the generated synthesis may remain fluent and factually plausible while being **incomplete**. Unlike obvious hallucinations, omissions are difficult to detect because the missing information is not represented in the answer.

4.2 Cross-document comparison

Research synthesis frequently requires comparing findings across multiple studies. Consider a literature review containing 100 papers. One paper may report a positive effect, another a null result, and a third may identify a methodological confound that explains the difference.

A model that independently summarizes each paper may perform adequately, but integrating the findings requires establishing relations between evidence separated across the context. LooGLE's findings concerning long dependencies illustrate why this remains challenging even when the document technically fits inside the model's context window.

The problem can be represented conceptually as:

[S = f(E_1,E_2,,E_n,R)]

where (E_i) represents evidence from source (i), (R) represents the research question, and (S) is the resulting synthesis.

In a robust system, the contribution of each relevant (E_i) should depend primarily on evidential relevance and quality rather than its position in the prompt. Under context degradation, however, the effective contribution can be approximated as:

[E_i' = E_i w(p_i,L,C)]

where (p_i) is evidence position, (L) is total context length, and (C) represents contextual complexity. If (w) varies substantially with position or length, identical evidence can receive different effective influence depending on how the prompt is constructed.

4.3 Contradiction detection

Scholarly synthesis requires more than accumulating supporting statements. It must identify disagreement.

Contradictions may occur because studies:

- use different populations;
- operationalize variables differently;
- employ different statistical models;
- measure outcomes at different times;
- have different sample sizes;
- report different effect estimates; or
- reach different conclusions.

Long-context degradation can make such contradictions harder to identify because the evidence needed for comparison may be distributed across distant sections. A model may generate a coherent consensus narrative even when the literature is genuinely divided.

4.4 Provenance and citation fidelity

A reliable research synthesis must preserve the relationship between claims and sources. RAG was partly motivated by the need for explicit access to external knowledge and provenance.

Long-context processing complicates provenance because the model may integrate information from multiple sources without clearly distinguishing which source supports which statement. The longer the input, the more difficult it becomes to audit whether every generated claim is adequately supported.

This is particularly important in academic writing, where a synthesis that sounds authoritative but cannot establish claim-level provenance is methodologically weak.

4.5 Abstraction and compression

Research synthesis necessarily involves abstraction: hundreds of pages must become a small number of arguments. Yet compression can eliminate precisely the qualifications that distinguish reliable scholarly reasoning from oversimplification.

For example, a study's main conclusion may be positive, while its limitations section indicates that the result applies only to a narrow population. A compression system that retains the conclusion but removes the qualification has produced a semantically plausible but scientifically distorted representation.

Thus, **compression is not information-neutral**. Its quality must be evaluated in terms of downstream reasoning, not merely similarity between compressed and original text.

5. Current Approaches and Developments

5.1 Long-context model training

One major strategy is to train models directly on longer sequences. LongT5 demonstrated that specialized architectures and pretraining can improve long-document summarization and question answering.

More recent work has extended this principle to substantially larger context lengths. LongRoPE reported extension of pretrained models toward multi-million-token contexts using positional interpolation and progressive training. Other approaches focus on specialized long-context data augmentation and training strategies. Tian et al. (2025), for example, reported strong RULER performance from an efficient long-context data augmentation strategy.
Such developments are promising, but their results should be interpreted carefully: improvements on synthetic retrieval benchmarks do not necessarily imply equivalent improvements in scholarly synthesis.

5.2 Retrieval-augmented generation

RAG remains one of the most practical solutions for large research corpora. Rather than inserting every document into the prompt, the system retrieves evidence relevant to a query and passes a smaller subset to the generator.

Its principal advantage is **context selectivity**. A research system can retrieve evidence at multiple levels:

1. document-level retrieval;
2. section-level retrieval;
3. passage-level retrieval;
4. claim-level retrieval; and
5. iterative retrieval based on intermediate synthesis.

However, retrieval can create a "retrieval bottleneck." If relevant evidence is not retrieved, a highly capable generator cannot use it. Conversely, over-retrieval can recreate the original long-context problem.

The most promising direction is therefore not maximum retrieval but **adaptive evidence selection**.

5.3 Prompt and context compression

Prompt compression attempts to preserve the information necessary for downstream reasoning while reducing context length.

LongLLMLingua uses question-aware coarse-to-fine compression and document reordering. Jiang et al. (2024) report performance gains in long-context settings together with substantial reductions in token usage and latency.

More recent approaches continue this trajectory. BRIEF-Pro, reported in 2026, uses context compression to distill relevant evidence from contexts exceeding 10,000 words into concise representations for multi-hop reasoning. The authors report improved QA performance under high compression ratios compared with LongLLMLingua while reducing computational overhead.

The challenge is that compression introduces a new optimization

problem: [_{C'} ; Q(C',R) |C'| |C|]

where (C) is the original context, (C') is the compressed context, (R) is the research task, and (Q) measures downstream synthesis quality.
The crucial criterion is therefore not whether the compressed text resembles the original but whether it preserves **decision-relevant evidence**.

5.4 Document reordering and position engineering

If model performance varies with evidence position, deliberately placing important material in favorable positions can improve performance. LongLLMLingua incorporates document reordering as one component of its long-context strategy.

This is an inexpensive intervention, but it should not be regarded as a complete solution. If a synthesis requires hundreds of independent evidence items, only a limited number can occupy the beginning or end of a context. Furthermore, reordering may itself introduce bias if the system systematically prioritizes evidence according to heuristics rather than methodological quality.

5.5 External and persistent memory

Memory architectures attempt to avoid treating the entire history of information as a single context window. LongMem, for example, combines a frozen backbone language model with a separate memory mechanism capable of retrieving information from an effectively unbounded memory bank. Wang et al. (2023) demonstrated improvements on long context modeling and memory-augmented in-context learning tasks.

For scholarly research, this suggests a potentially useful architecture in which the system maintains structured representations of:

- documents;
- claims;
- methods;
- findings;
- limitations;
- citations;
- contradictions; and
- evidence provenance.

Such a memory could be queried iteratively rather than reconstructed from raw documents at every generation step.

5.6 Position-agnostic training

He et al. (2024) proposed position-agnostic decompositional training to improve long context QA. Their experiments report substantial gains under shuffled settings, indicating that targeted training can reduce sensitivity to evidence position.

This line of work is important because it treats context degradation as a learnable behavioral problem rather than merely an architectural limitation.

6. Research Challenges and Limitations

6.1 Context capacity versus effective context

The central methodological challenge is defining and measuring **effective context length**. A model may accept one million tokens while reliably reasoning over only a much smaller subset.

Future evaluations should therefore report performance curves as a function of context length rather than a single maximum context value.

6.2 Benchmark realism

Many long-context benchmarks rely on synthetic retrieval tasks. Such tests are valuable for isolating specific capabilities, but scholarly synthesis is considerably more complex. Real research requires:

- heterogeneous documents;
- ambiguous questions;
- competing explanations;
- source-quality assessment;
- citation tracking;
- quantitative evidence;
- uncertainty;
- temporal differences; and
- domain-specific terminology.

LongBench provides broader task coverage, while LooGLE emphasizes long dependencies, but neither fully captures the methodological complexity of scholarly literature review.

6.3 Summarization versus synthesis

A major evaluation problem is conflating summarization with synthesis. Summarization can be evaluated against a source document, whereas synthesis requires reasoning across multiple sources.

A high-quality synthesis should be evaluated for:

[Q_{} = f(A,C,P,I,U)]

where:

- (A) = factual accuracy;
- (C) = coverage;
- (P) = provenance fidelity;
- (I) = cross-source integration; and
- (U) = uncertainty calibration.

Existing benchmarks often emphasize only one or two of these

dimensions.

6.4 Compression and information loss

Compression is inherently lossy. A system must decide which information to discard before it knows all possible downstream uses of that information.

This creates a particularly serious problem in open-ended research. A detail that appears irrelevant during the initial question may later become essential when a contradictory study is discovered.

6.5 Model and benchmark instability

Long-context evaluations are highly sensitive to model version, tokenizer, context formatting, retrieval configuration, and prompt structure. Consequently, claims that one method “solves” long-context degradation should be interpreted cautiously unless replicated across models and domains.

Recent evidence from long-context research also shows that increasing context capacity can involve trade-offs with shorter-context performance and substantial training costs.

7. Research Gaps and Future Directions

7.1 Evidence-centric long-context architectures

Future systems should move from token-centric context management toward **evidence centric representations**. Rather than storing research solely as sequences of tokens, systems could represent each source through structured units such as:

[{,,,,}.]

The generator could then retrieve and compare these units according to the research question.

7.2 Iterative synthesis rather than single-pass synthesis

A more reliable workflow would be:

1. identify the research question;
2. retrieve candidate sources;
3. extract claims and evidence;
4. cluster related findings;
5. identify contradictions;
6. assess source quality;
7. synthesize intermediate findings;
8. retrieve additional evidence for unresolved questions;
9. perform a contradiction and provenance audit; and
10. generate the final synthesis.

This architecture treats research synthesis as an iterative reasoning process rather than a single generation call.

7.3 Claim-level provenance

Future models should explicitly connect every substantive synthesis claim to its supporting evidence. This could be implemented as a graph:

[.]

Such representations would make academic outputs more auditable and reduce the risk of unsupported generalizations.

7.4 Contradiction-aware synthesis

Research systems should explicitly search for counterevidence. Instead of asking only "What does the literature say?", systems should separately ask:

- Which studies disagree?
- Why do they disagree?
- Are the populations different?
- Are the measurements different?
- Are methodological weaknesses responsible?
- Does the evidence justify a consensus?

This approach could reduce the tendency of generative models to transform heterogeneous evidence into overly coherent narratives.

7.5 Dynamic context allocation

Rather than allocating a fixed context budget, future systems could dynamically determine how much context each evidence unit deserves. A high-value passage might receive repeated or expanded processing, while low-value material could remain compressed.

This suggests an architecture combining retrieval, memory, compression, and iterative reasoning rather than selecting one technique as a universal solution.

7.6 Long-context evaluation for genuine scholarship

A major research opportunity is the development of benchmarks based on real scholarly corpora. Such benchmarks should evaluate:

- literature-review completeness;
- evidence attribution;
- citation correctness;
- contradiction detection;
- methodological comparison;
- temporal reasoning;
- quantitative result extraction;
- uncertainty representation; and
- resistance to irrelevant-context interference.

The benchmark should also measure performance as a function of context length and evidence position. This would transform "long context" from a model marketing specification into an empirically testable capability.

7.7 Human–AI collaborative synthesis

Finally, long-context research systems should not necessarily seek complete automation. Human researchers are particularly valuable for determining whether methodological differences invalidate an apparent comparison and whether a synthesis appropriately represents disciplinary debates.

A useful architecture may therefore treat the model as an **evidence organizer and synthesis assistant**, while maintaining human oversight for high-level interpretation and epistemic judgment.

8. Conclusion

The expansion of LLM context windows represents an important technological development, but it does not eliminate the fundamental difficulty of reasoning over large quantities of information. Empirical research demonstrates that models can exhibit positional biases, lose information located in the middle of long contexts, and experience performance degradation merely as input length increases.

For long-document research synthesis, these limitations are particularly consequential. Scholarly synthesis depends not only on retrieving facts but also on determining which evidence matters, integrating information across sources, detecting contradictions, preserving provenance, and calibrating conclusions to the strength of the evidence. A model can therefore produce a fluent synthesis while silently omitting relevant studies, underweighting qualifications, or collapsing methodological disagreement.

The literature suggests that no single intervention is sufficient. Long-context training and positional-encoding methods increase capacity; retrieval reduces unnecessary context; compression increases information density; reordering can mitigate positional effects; memory architectures externalize persistent information; and iterative reasoning can improve cross-document integration. Recent work indicates that these approaches are increasingly being combined rather than treated as mutually exclusive alternatives.

The central research challenge is therefore to move from **longer context windows to more reliable effective contexts**. For scholarly applications, the goal should not be to maximize the number of tokens processed by a model, but to maximize the amount of *relevant, attributable, correctly integrated evidence* that survives the research-synthesis process.

In this sense, context-window degradation is not merely an engineering inconvenience. It is an epistemic problem. If long-context systems are to become dependable research instruments, future work must evaluate them according to the standards of scholarship— coverage, evidence fidelity, methodological sensitivity, provenance, contradiction awareness, and uncertainty—rather than according to context length alone.